organisms, ecosystems are open systems, absorbing energy and mass and releasing heat and waste products.

Most gains and losses to ecosystems are small compared to the amounts that cycle within them. Even so, the balance between inputs and outputs is important because it determines whether an ecosystem stores or loses a given element. In particular, if a nutrient's outputs exceed its inputs, that nutrient will eventually limit production in that ecosystem. Human activities often change the balance of inputs and outputs considerably, as we'll see later in this chapter and in Concept 56.4.

Energy, Mass, and Trophic Levels

Ecologists group species in an ecosystem into trophic levels based on feeding relationships (see Concept 54.2). The trophic level that ultimately supports all others consists of autotrophs, also called the **primary producers** of the ecosystem. Most autotrophs are photosynthetic organisms that use light energy to synthesize sugars and other organic compounds, which they use as fuel for cellular respiration and as building material for growth. The most common autotrophs are plants, algae, and photosynthetic prokaryotes. However, chemosynthetic prokaryotes are the primary producers in some ecosystems, such as deep-sea hydrothermal vents (see Figure 52.16) and places deep under the ground or ice.

Organisms in trophic levels above the primary producers are heterotrophs (consumers), which depend directly or indirectly on the outputs of primary producers for their source of energy. Herbivores, which eat plants and other primary producers, are **primary consumers**. Carnivores that eat herbivores are **secondary consumers**, and carnivores that eat other carnivores are **tertiary consumers**.

Another group of heterotrophs is the **decomposers**, consumers that get their energy from detritus. **Detritus** is nonliving organic material, such as the remains of dead organisms, feces, and fallen leaves. Although some animals (such as earthworms) feed on detritus, the main decomposers are prokaryotes and fungi **(Figure 55.3)**. These organisms secrete enzymes that digest organic material; they then absorb the breakdown products. Many decomposers are in turn eaten by secondary and tertiary consumers. In a forest, for instance, birds eat earthworms that have been feeding on leaf litter and its associated prokaryotes and fungi. As a result, chemicals originally synthesized by plants pass from the plants to leaf litter to decomposers to birds.

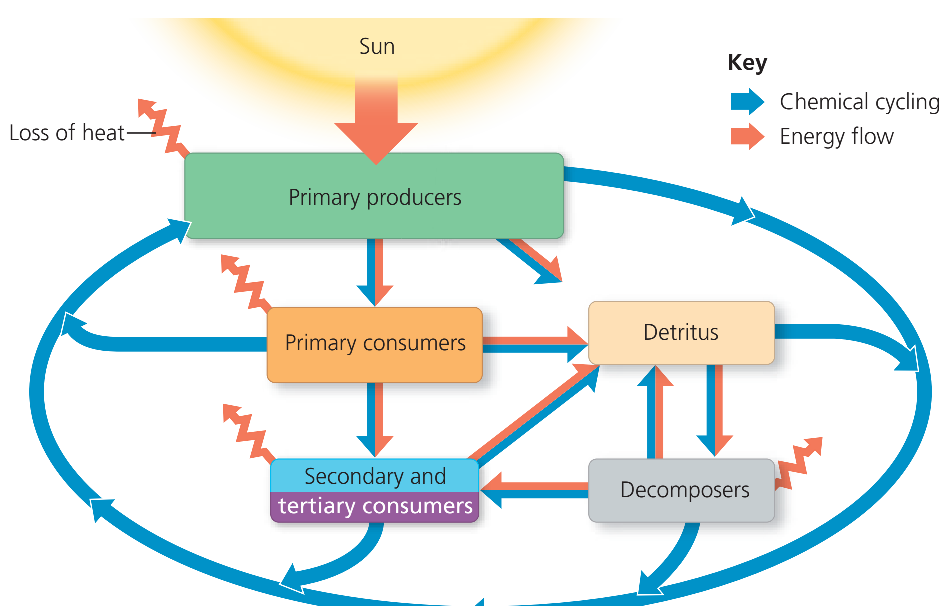
By recycling chemical elements to producers, decomposers also play a key role in the trophic relationships of an ecosystem **(Figure 55.4)**. Decomposers convert organic matter from all trophic levels to inorganic compounds usable by primary producers. When the decomposers excrete waste products or die, those inorganic compounds are returned to the soil. Producers can then absorb these elements and use them to synthesize

▼ Figure 55.3 Decomposers.

▲ Rod-shaped and spherical bacteria in compost (colorized SEM)

▼ Fungi decomposing a dead tree

▶ Figure 55.4 An overview of energy and nutrient dynamics in an ecosystem. Energy enters, flows through, and exits an ecosystem, whereas chemical nutrients cycle within it. Energy (dark orange arrows) entering from the sun as radiation is transferred as chemical energy through the food web; each of these units of energy ultimately exits as heat radiated into space. Most transfers of nutrients (blue arrows) through the food web lead eventually to detritus; the nutrients then cycle back to the primary producers.



VISUAL SKILLS *In this diagram, one blue arrow leads to the box labeled "Primary consumers," and three blue arrows come out of it. For each of these arrows, describe an example of nutrient transfer that the arrow could represent, using specific organisms and other components of an ecosystem in your answer.*

Mastering Biology Figure Walkthrough